arXiv:2611.12426v1 [cs.CV] 9 Nov 2026

LongCat

dNaViT: Discrete Codes are General-Purpose Visual Representations

Meituan LongCat Team

Discrete autoregressive modeling, notably through next-token prediction, has become a cornerstone of modern language models. Extending this paradigm to vision allows visual and language models to converge on a unified framework, thereby fostering shared methodology and infrastructure. This work investigates a central question: how far can a finite discrete vocabulary go in representing the richness of the visual world? We present dNaViT, a Discrete Native-resolution Vision Transformer that produces discrete visual primitives analogous to language tokens. Built upon a frozen pretrained encoder, dNaViT applies lightweight residual modulation with a dual-branch objective to inject reconstruction-aware cues into semantic features. By leveraging grouped product quantization, it scales discrete capacity without inflating spatial sequence lengths, while a balanced training recipe maintains both semantic and pixel-level information. Consequently, dNaViT functions as a versatile discrete visual tokenizer, enabling unified encoding across downstream tasks alongside high-fidelity image decoding. Extensive evaluations show that dNaViT matches or exceeds task-specific baselines across 12 visual understanding benchmarks, 4 generation benchmarks, and an image editing benchmark. These findings demonstrate that a compact discrete vocabulary can serve as a universal visual currency and unlock the full potential of fully unified, native multimodal methodology.

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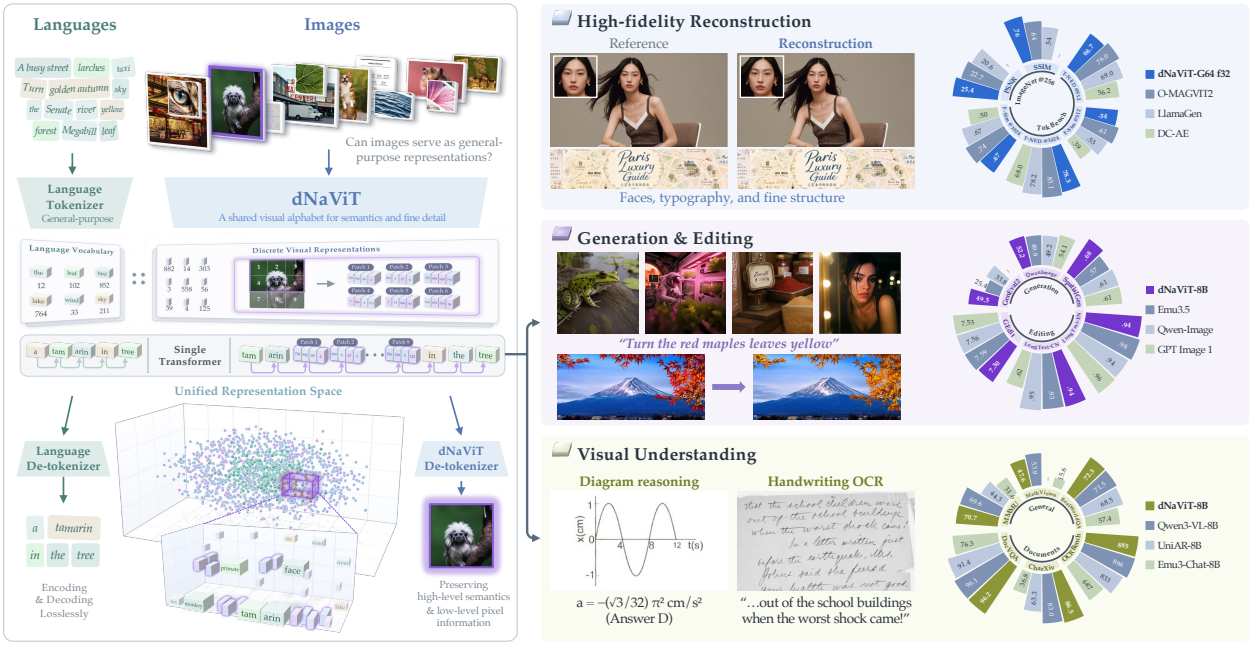


Figure 1 dNaViT: A unified visual tokenizer for general-purpose modeling. dNaViT represents arbitrary-resolution images as discrete visual primitives, natively unifying *reconstruction, understanding, generation, and editing* under a shared vocabulary. By achieving performance competitive with specialized baselines across all domains, dNaViT validates a shared token-based paradigm for vision, analogous to language modeling.

Correspondence: lixiaotong.edu@gmail.com

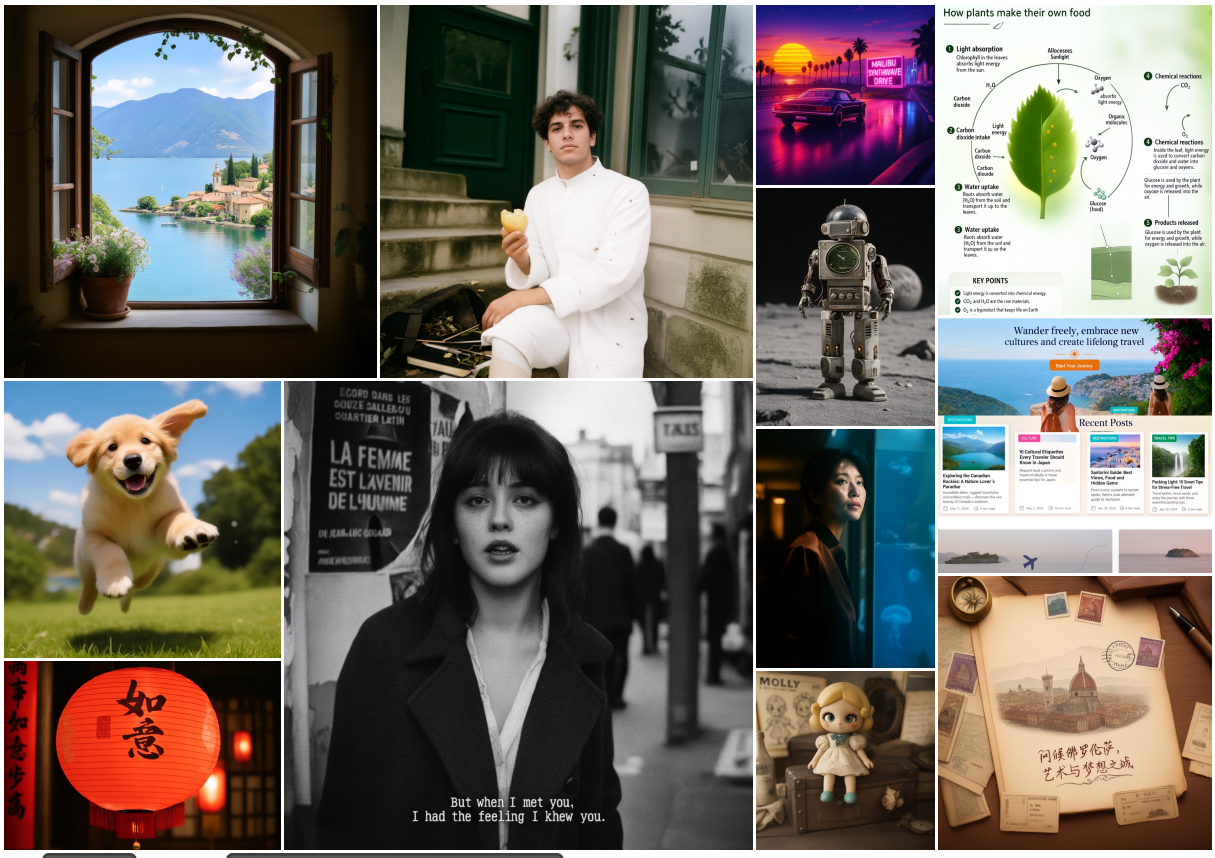


Figure 2 dNaViT's unified discrete representation as a foundation for diverse downstream visual tasks. Qualitative showcases illustrate the versatility of this shared representation.

Contents

1 Introduction **4**

2 dNaViT **5**
- 2.1 Background . . . 5
- 2.2 Overall Architecture . . . 6
- 2.3 Representation Encoder . . . 6
- 2.4 Quantizer . . . 7
- 2.5 Dual-Branch Reconstruction . . . 8
- 2.6 Refiner . . . 9

3 Implementation **10**
- 3.1 Training Stage . . . 10
- 3.2 Training Data . . . 12

4 Experiment **12**
- 4.1 Image Reconstruction . . . 12
- 4.2 Image Classification . . . 14
- 4.3 Visual Understanding . . . 16
- 4.4 Visual Generation and Editing . . . 18

5 Analysis **18**
- 5.1 Effectiveness of ResMod . . . 18
- 5.2 Comparison between Discrete and Continuous Approaches . . . 19
- 5.3 Comparison of Pixel Reconstruction . . . 19
- 5.4 Qualitative Visual Understanding Examples . . . 20

6 Conclusion **20**

7 Contributors **20**

1 Introduction

Language models have achieved remarkable success through next-token prediction, anchored by discrete tokens as their fundamental modeling unit (Brown et al., 2020). From the perspective of *compression as intelligence* (Delétang et al., 2023), discrete tokens provide a unified substrate for representation learning and autoregressive modeling. In contrast, modern multimodal models frequently rely on modality-specific representations and heterogeneous learning objectives (Caffagni et al., 2024). This paradigm divergence disconnects vision from the established techniques and scalable infrastructure to language, raising a foundational question: can visual information be represented losslessly like language using a learned discrete vocabulary?

Discrete visual representations offer a promising bridge, naturally aligning visual inputs with autoregressive language architectures under a unified token-prediction objective (Xie et al., 2024; Wang et al., 2024b; Zou et al., 2026b; Peng et al., 2026; Meituan LongCat Team et al., 2026; Cui et al., 2025). However, discretizing dense, continuous visual signals into a finite vocabulary presents a severe information bottleneck (Ma et al., 2025; Yu et al., 2026) that hinders performance. Crucially, harmonizing high-level semantic abstraction for understanding with fine-grained pixel details for high-fidelity reconstruction within a single discrete bottleneck remains an unresolved tension (Wu et al., 2024; Qu et al., 2025).

To answer the question posed above and address these challenges, we introduce **dNaViT**, a Discrete Native-resolution Vision Transformer that extends our preliminary work in LongCat-Next (Meituan LongCat Team et al., 2026). Following the representation encoder paradigm (Zheng et al., 2025), dNaViT enriches the semantic features of a frozen pretrained encoder with pixel details through a *Residual Modulator* (ResMod), without compromising its underlying semantic capability. The resulting features are discretized into a shared vocabulary using Grouped Product Quantization (PQ) (Jégou et al., 2011) under dual-branch representation and pixel reconstruction loss. Training progresses through continuous initialization, discrete feature alignment, and final joint optimization, which balance theses objectives and suppress gradient interference.

These design choices yield a shared discrete interface for tasks that conventionally rely on separate visual encodings, extending the utility of pretrained representations beyond their original objectives. By rotating and translating pretrained features, residual modulation reorganizes their geometry to accommodate reconstruction-relevant information, making more effective use of the available representational capacity. Combined with sufficient factorized code capacity, this allows semantic structure and fine-grained detail to coexist within a shared discrete bottleneck, mitigating information loss during quantization. The resulting tokenizer brings *the flexibility of text tokenization* to vision: images can be encoded into discrete sequences and decoded into faithful reconstructions, while the codes retain semantic information closely aligned with language.

We evaluate the generality of dNaViT across four core capabilities: image reconstruction, classification, multimodal understanding, generation, editing. Driven by a single shared discrete vocabulary with a spatial downsampling factor $f = 32$, dNaViT achieves 28.12 dB PSNR and 0.81 SSIM at 512×512 resolution without sacrificing semantic representation capability. When paired with Qwen3-8B, dNaViT surpasses prior discrete baselines and rivals Qwen3-VL-8B across key document and reasoning benchmarks, scoring 885 on OCRBench. For image generation and editing, dNaViT achieve impressive performance on GenEval2 (49.5) and outperforms open-weight baselines on Qwen-Image-Bench (52.2), while excelling at bilingual text rendering. Crucially, dNaViT establishes these comprehensive understanding and generation capabilities under a substantially reduced data budget relative to existing models.

These findings suggest that the gap between continuous visual signals and discrete tokens is a challenge of representation learning rather than an inherent limitation. With appropriate modeling, discrete codes can serve as general-purpose visual primitives, bringing vision and language into a shared token-based paradigm. Our main contributions are summarized as follows:

1. This work explores the question of *whether discrete codes can serve as general-purpose visual primitives*, analogous to the role of tokens in language modeling. To investigate this possibility, we introduce dNaViT, a native-resolution visual tokenizer designed to capture both high-level semantic structure and fine-grained pixel information within a shared vocabulary for different tasks.
2. dNaViT uses residual modulation to aggregate multi-level features from a frozen vision encoder and grouped product quantization to encode the enriched representations. Progressive training coordinates dual-branch

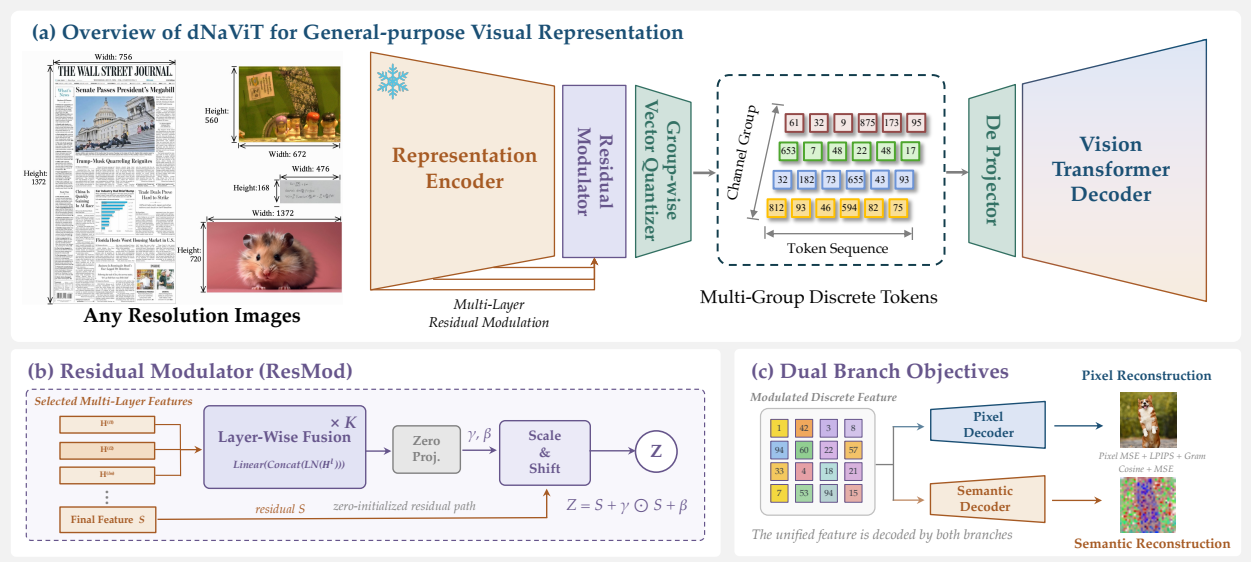


Figure 3 Overview of dNaViT. (a) A frozen representation encoder handles arbitrary resolutions, with ResMod injecting multi-level information before group-wise product quantization. The shared discrete codes are decoded by individual ViT decoders into pixel and semantic reconstructions. (b) ResMod fuses intermediate features and predicts a zero-initialized scale-and-shift residual. (c) Pixel and semantic branches are optimized with image and feature reconstruction losses, respectively.

semantic and pixel reconstruction objectives. Experiments demonstrate that this design balances semantic preservation and reconstruction fidelity within the discrete bottleneck.

3. Comprehensive evaluations across visual understanding, image reconstruction, generation, and editing demonstrate dNaViT's generality. With a substantially smaller training data budget than the compared models, dNaViT achieves strong understanding and generation performance, highlighting the effectiveness and data efficiency of its unified, language-aligned representation.
4. This work demonstrates that a shared discrete token space can serve as a *universal visual currency*, providing a general-purpose intermediate representation across complementary tasks and a foundation for fully unified, native multimodal modeling.

2 dNaViT

This section presents dNaViT, a visual tokenizer that learns shared discrete codes as a unified interface for diverse downstream tasks, replacing separate task-specific feature representations. We will describe how the architecture supports these capabilities while mitigating conflicts between semantic abstraction and fine-grained detail preservation, as well as information loss during discretization.

2.1 Background

Visual modeling has largely relied on distinct continuous representations for different downstream tasks: pretrained semantic features for multimodal understanding (Tschannen et al., 2025) and reconstruction-oriented latents for diffusion-based synthesis (Rombach et al., 2022). Representation autoencoders bridge these approaches by pairing pretrained semantic encoders with reconstruction decoders (Zheng et al., 2025). By contrast, language models use discrete tokens as a uniform representation of text, supporting a unified, scalable next-token prediction framework. Meanwhile, recent benchmarks (An et al., 2026a; Yan et al., 2026) highlight the growing need to integrate linguistic and visual intelligence to solve real-world tasks.

Extending this framework to vision is challenging: unlike the discrete symbols of language, visual signals encode dense, continuous variation. Discrete visual tokenization addresses this by mapping images to sequences of codebook indices. For example, vector-quantized autoencoders assign continuous features to learned codebook

entries and reconstruct images from the corresponding embeddings (van den Oord et al., 2017). These indices can be embedded and predicted like text tokens, enabling a shared autoregressive interface and the reuse of language-model architectures and objectives. Chameleon (Chameleon Team, 2024) and Emu3 (Wang et al., 2024b) demonstrate this formulation for multimodal understanding and generation. However, information loss during tokenization can constrain downstream performance and the benefits of scaling.

A shared token interface alone does not guarantee that the learned codes jointly preserve semantic information and fine-grained visual detail, both of which are essential to general-purpose visual modeling. Prior studies have examined the distinct information requirements of understanding and reconstruction (Qu et al., 2025) and identified limited discrete capacity as a key bottleneck in supporting both objectives (Ma et al., 2025). Motivated by these findings, dNaViT aims to preserve complementary task-relevant information within a shared discrete vocabulary and match the performance of task-specific representations.

2.2 Overall Architecture

Motivated by the exploration of information recovery through residual connections (Meituan LongCat Team et al., 2026), dNaViT further propose residual modulation to aggregate complementary cues and enrich the features of a frozen native-resolution vision encoder with reconstruction-relevant detail (Fig. 3). To preserve both semantic structure and visual detail within a shared discrete bottleneck, we quantize the enriched features using grouped product quantization and supervise the resulting codes through dual reconstruction branches. Progressive training coordinates these objectives to mitigate interference during optimization.

2.3 Representation Encoder

Unified visual tokenization must reconcile appearance invariance for understanding with sensitivity to detail for reconstruction. RAEs demonstrate reconstruction from continuous semantic features (Zheng et al., 2025), while LongCat-Next suggests that residual connections can carry low-level information across a semantic encoder (Meituan LongCat Team et al., 2026). This motivates us to directly leverage complementary cues by recombining multiple encoder depths, without even training the encoder itself. Meanwhile, quantization may collapse weakly expressed appearance differences into the same code. It therefore requires retaining the final semantic representation as an explicit anchor throughout quantization to preserve representation quality.

To tackle this, we introduce a *Residual Modulator* (ResMod) inspired by frozen semantic base with detail-aware modulation of PAE (Yue et al., 2026), which fuses intermediate features from the frozen encoder and learns a spatially varying correction to its final representation. Multi-level fusion exposes complementary spatial cues, and a zero-initialized residual update preserves the pretrained features at initialization. The resulting feature grid provides the quantizer with a shared representation whose semantic structure and local detail can be supervised by the two reconstruction branches.

Residual Modulator. Figure 3(b) illustrates the module. Let $\{\mathbf{H}^{(\ell)}\}_{\ell=0}^{L}$ be the spatial tokens of the frozen vision transformer, with $\mathbf{H}^{(\ell)} \in \mathbb{R}^{N\times D}$. We select early and deeper layers $\mathcal{S} \subseteq \{0,\dots,L\}$ and fuse their normalized features:

$$\mathbf{U} = \mathcal{T}_\theta\Big(\mathcal{F}_\theta\Big(\mathrm{Concat}_{\ell\in\mathcal{S}}\Big[\mathrm{LN}_\ell(\mathbf{H}^{(\ell)})\Big]\Big)\Big), \tag{1}$$

where $\mathcal{F}_\theta$ projects the concatenated features to a common width and $\mathcal{T}_\theta$ is a lightweight stack of self-attention and feed-forward blocks. These intermediate taps provide direct access to spatial information at different levels of abstraction.

Let $\mathbf{S} = \mathcal{A}(\mathbf{H}^{(L)})$ denote the semantic base on the quantizer grid, where $\mathcal{A}$ is the identity or the encoder's spatial aggregation operator. We write N_q and C for the number of spatial positions and the channel width on this grid, respectively. A projection $\mathcal{P}_\theta$ aligns $\mathbf{U}$ to this grid, and an affine head predicts token-wise scale and shift parameters:

$$(\boldsymbol{\gamma},\boldsymbol{\beta}) = \mathrm{Split}\big(\mathcal{W}_\theta\big(\mathrm{LN}(\mathcal{P}_\theta(\mathbf{U}))\big)\big), \qquad \mathbf{Z} = \mathbf{S} + \underbrace{\boldsymbol{\gamma}\odot\mathbf{S}+\boldsymbol{\beta}}_{\boldsymbol{\Delta}_\theta}. \tag{2}$$

The projection $\mathcal{P}_\theta$ also aggregates spatial tokens when $\mathbf{S}$ uses a coarser grid. We initialize $\mathcal{W}_\theta$ to zero, giving $\boldsymbol{\Delta}_\theta = \mathbf{0}$ and $\mathbf{Z} = \mathbf{S}$ at initialization. Training then learns how to reweight the semantic base and add

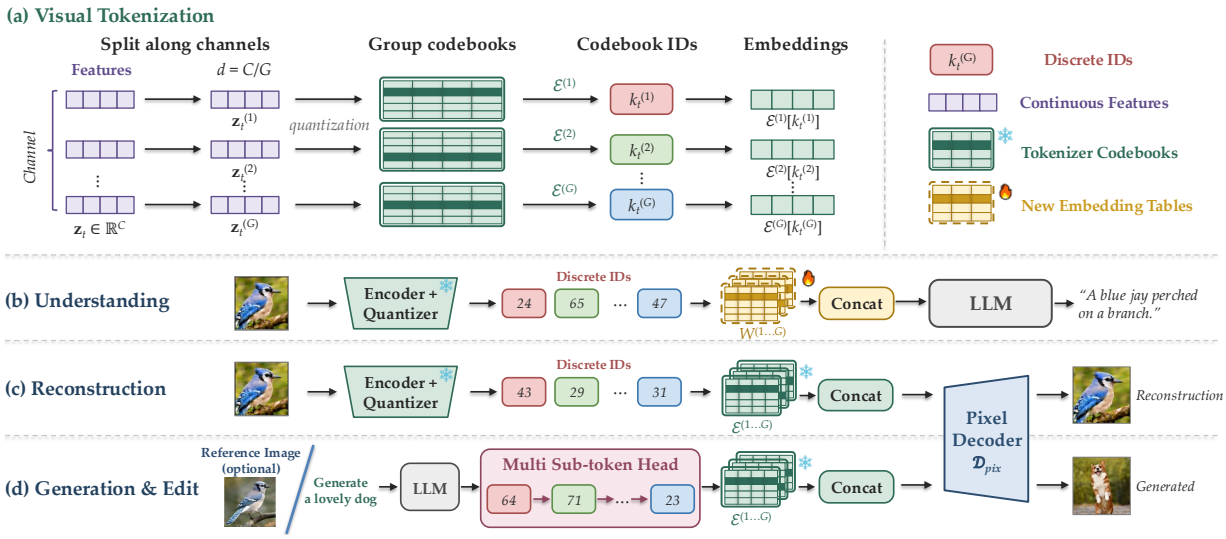


Figure 4 Tokenization and Unified Codes for Downstream Tasks. (a) At spatial position t, the modulated feature $\mathbf{z}_t$ is split into G channel groups and quantized with group-specific codebooks $\mathcal{E}^{(g)}$, yielding indices $(k_t^{(1)}, \ldots, k_t^{(G)})$ and their embeddings. (b) Understanding supplies only code indices to the language model (LM) through newly initialized group-wise embedding tables $\{\mathbf{W}^{(g)}\}_{g=1}^{G}$, trained jointly with the LM. (c) Reconstruction concatenates the indexed embeddings from the frozen tokenizer codebooks and feeds them to the pixel decoder $\mathcal{D}_{\text{pix}}$. (d) Generation and editing use a multi-sub-token head atop the LM to predict each position's multi-level indices, optionally conditioned on a reference image, then decode through the same frozen codebooks and pixel decoder.

complementary information at each position. Thus, residual pathways within the encoder provide access to information across depth, and ResMod organizes this information for the discrete bottleneck while keeping the pretrained encoder fixed. As a result, ResMod only translates and rotates the pretrained features to enlarge representation capacity while preserving their original semantic structure.

2.4 Quantizer

The residual modulator enriches the features entering the discrete bottleneck, but retaining the exposed semantic distinctions and spatial detail requires sufficient quantization capacity at each spatial position. Motivated by evidence that multi-codebook quantization with a sufficient bit budget preserves semantic information at a level competitive with continuous representations (Zou et al., 2026a), we adopt grouped product quantization (Jégou et al., 2011) to expand this capacity while keeping the spatial token grid unchanged. As illustrated in Fig. 4, the features are factorized along the channel dimension and encoded using multiple codebooks, avoiding a prohibitively large monolithic vocabulary.

Let $\mathbf{z}_t \in \mathbb{R}^C$ denote the feature presented to the quantizer at position t of the modulated representation $\mathbf{Z} \in \mathbb{R}^{N_q \times C}$. We partition its channels into G disjoint groups, $\mathbf{z}_t = \text{Concat}_{g=1}^{G}\left[\mathbf{z}_t^{(g)}\right]$, where each group has dimension $d = C/G$. Each group is assigned its own learned codebook $\mathcal{E}^{(g)} = \{\mathbf{e}_k^{(g)}\}_{k=1}^{K}$, with $\mathbf{e}_k^{(g)} \in \mathbb{R}^d$. Quantization selects the nearest entry within each group and concatenates the selected embeddings:

$$
\begin{aligned}
k_t^{(g)} &= \underset{k \in \{1,\ldots,K\}}{\arg\min} \left\| \mathbf{z}_t^{(g)} - \mathbf{e}_k^{(g)} \right\|_2^2, \\
\mathbf{k}_t &= (k_t^{(1)}, \ldots, k_t^{(G)}), \qquad \mathbf{c}_t = \text{Concat}_{g=1}^{G}\left[\mathbf{e}_{k_t^{(g)}}^{(g)}\right].
\end{aligned}
\tag{3}
$$

Here $\mathbf{k}_t$ is the discrete index tuple, whereas $\mathbf{c}_t$ is its codebook embedding supplied to downstream decoders. Codebooks are shared across spatial positions, so the same quantization rule applies to variable-length grids at different image resolutions and aspect ratios. The resulting tuple belongs to the Cartesian product

$\mathcal{K} = \{1, \ldots, K\}^G$, giving a combinatorial code space with

$$|\mathcal{K}| = K^G, \qquad B_{\text{position}} = G \log_2 K, \qquad B_{\text{image}} = N_q G \log_2 K, \tag{4}$$

where N_q denotes the number of spatial positions on the quantizer grid. With G codebooks of K entries each, the quantizer admits up to K^G combinations without explicitly storing a vocabulary of that size. At fixed K, increasing G raises the nominal bit budget while leaving N_q unchanged, but introduces more code indices to model. This budget characterizes the available discrete alphabet, not the empirical entropy or semantic information retained; greater capacity alone does not guarantee faithful reconstruction. Separate codebook lookups also do not imply statistically independent indices, as the groups encode the same visual content.

The quantization groups operate on the fused representation without predefined semantic or perceptual roles. Stacking the embeddings in spatial order gives $\mathbf{C} = [\mathbf{c}_1, \ldots, \mathbf{c}_{N_q}]^\top \in \mathbb{R}^{N_q \times C}$. The resulting matrix serves as a shared input to the semantic and pixel decoders, whose complementary objectives encourage the codes to retain both pretrained semantic structure and reconstruction-relevant detail. Residual modulation thus determines which information enters the bottleneck, grouped quantization supplies the capacity to encode it, and dual-decoder supervision guides how that capacity is used.

2.5 Dual-Branch Reconstruction

A general-purpose visual vocabulary must preserve pretrained semantic structure while retaining the fine-grained visual detail needed to recover the original visual signal. These requirements concern different output spaces but must be satisfied by the same discrete codes. We therefore introduce two specialized decoders that share the quantized representation. The semantic branch aligns the codes with continuous visual features, while the pixel branch supervises image reconstruction.

Representation Reconstruction Branch. We take the quantized feature matrix $\mathbf{C}$, as defined in Section 2.4, as the input to both decoders. The semantic decoder $\mathcal{D}_{\text{sem}}$ maps the quantized embeddings back to the continuous feature space after residual modulation,

$$\widehat{\mathbf{Z}} = \mathcal{D}_{\text{sem}}(\mathbf{C}). \tag{5}$$

We supervise this prediction with a fixed post-ResMod target $\mathbf{Z}^\star$, combining feature regression with cosine alignment,

$$\mathcal{L}_{\text{sem}} = 1 - \cos\Big(\widehat{\mathbf{Z}}, \text{sg}(\mathbf{Z}^\star)\Big) + \lambda^{\text{sem}}_{\text{MSE}} \Big\|\widehat{\mathbf{Z}} - \text{sg}(\mathbf{Z}^\star)\Big\|_2^2, \tag{6}$$

where $\text{sg}(\cdot)$ stops gradients through the target path. The target is produced by a frozen ResMod, as specified in Section 3.1. We find that cosine loss alone can be small even for inaccurate reconstructions when a dominant shared feature mean concentrates features in a narrow cone, consistent with representation anisotropy (Ethayarajh, 2019). The MSE term complements directional alignment by constraining absolute feature values. Reconstructing this representation encourages the codes to retain the semantic structure inherited from the pretrained backbone together with the information introduced by residual modulation.

Pixel Reconstruction Branch. Feature alignment alone does not directly constrain fidelity to the original image. We therefore decode the same embeddings into pixels, first restoring and expanding their spatial grid through token unmerging,

$$\widehat{\mathbf{x}} = \mathcal{D}_{\text{pix}}\big(\mathcal{U}(\mathbf{C}; h_q, w_q)\big), \tag{7}$$

where (h_q, w_q) is the quantized grid size, $\mathcal{U}$ denotes token unmerging and spatial expansion, and $\mathcal{D}_{\text{pix}}$ is the pixel decoder. We train this branch with five complementary reconstruction losses,

$$\mathcal{L}_{\text{pix}} = \lambda_1 \mathcal{L}_1 + \lambda^{\text{pix}}_{\text{MSE}} \mathcal{L}_{\text{MSE}} + \lambda_{\text{LPIPS}} \mathcal{L}_{\text{LPIPS}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}} + \lambda_{\text{Gram}} \mathcal{L}_{\text{Gram}}. \tag{8}$$

The λ coefficients balance the five objectives; the pixel MSE weight $\lambda^{\text{pix}}_{\text{MSE}}$ is distinct from the feature MSE weight $\lambda^{\text{sem}}_{\text{MSE}}$. All terms compare the reconstruction $\widehat{\mathbf{x}}$ with the input $\mathbf{x}$. Here $\mathcal{L}_1$ and $\mathcal{L}_{\text{MSE}}$ denote mean absolute and mean squared pixel errors, respectively. $\mathcal{L}_{\text{LPIPS}}$ uses VGG features to measure perceptual similarity across multiple layers. The ConvNeXt-S-based perceptual loss $\mathcal{L}_{\text{perc}}$ applies MSE to the network

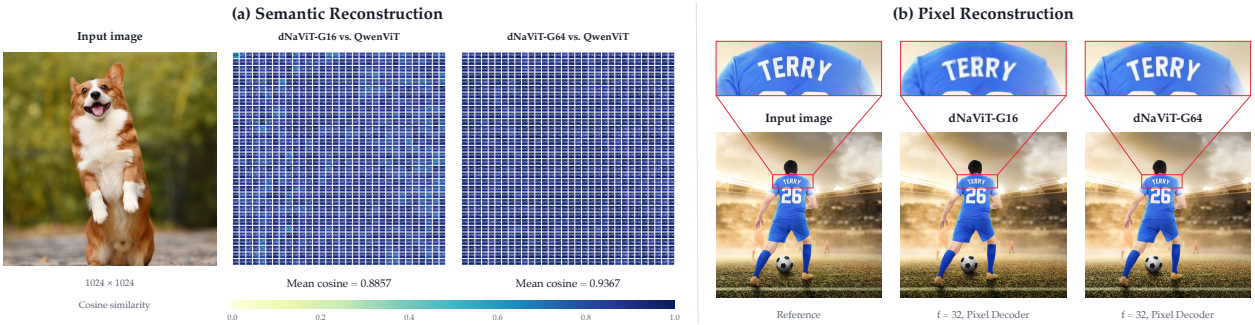


Figure 5 Joint preservation of high-level semantics and low-level pixel detail in a shared discrete representation. (a) dNaViT's reconstructed semantic features closely match the original QwenViT features. Each cell reports the cosine similarity between corresponding tokens. (b) Pixel reconstructions from dNaViT-G16 and dNaViT-G64, with matched crops highlighting the fine-grained text on the jersey.

outputs to align high-level recognition responses. $\mathcal{L}_{\text{Gram}}$ matches Gram matrices of VGG features to preserve texture and channel correlations. The two branches jointly constrain the information retained by the shared vocabulary through

$$\mathcal{L}_{\text{dual}} = \lambda_{\text{sem}}\mathcal{L}_{\text{sem}} + \lambda_{\text{pix}}\mathcal{L}_{\text{pix}}. \tag{9}$$

The semantic objective anchors reconstruction in the modulated feature space, while the pixel objective requires access to texture, color, and spatial detail. Section 3.1 describes how these objectives are introduced progressively to train the shared bottleneck and residual pathway.

Flexible Encoding and Decoding at Arbitrary Resolutions. We retain each image's token-grid dimensions throughout encoding and decoding. The encoder maps an input image to an $h_q \times w_q$ feature grid, which quantization converts into codebook embeddings without altering its spatial layout. For semantic reconstruction, the semantic decoder recovers a continuous feature at each grid position. For image reconstruction, $\mathcal{U}(\mathbf{C}; h_q, w_q)$ uses the recorded dimensions to arrange and spatially expand the quantized tokens before the pixel decoder reconstructs an RGB image at the input resolution. Both branches therefore remain spatially aligned with the input image across varying grid dimensions and aspect ratios.

2.6 Refiner

Although the ViT pixel decoder preserves image content faithfully, standard practices often employ a refiner module to further enhance its visual perceptual quality. We introduce a code-conditioned Refiner for high-fidelity decoding with sharper, more realistic details (Fig. 6). Unlike conventional super-resolution models conditioned on decoded RGB images, our Refiner directly conditions on discrete visual codes, avoiding information loss introduced by intermediate pixel decoding. It also requires no text conditioning, whose semantic priors may alter source content and constrain image editing. Built on a pretrained image-generation prior, the Refiner replaces the ViT pixel decoder with a flow decoder using discrete codes as its sole conditioning input. We train a multi-step teacher and distill it into an efficient one-step student.

Code-conditioned refinement. Given an encoder observation $\mathbf{x}_{\text{L}}$ and a target image $\mathbf{x}_{\text{H}}$, the frozen tokenizer maps $\mathbf{x}_{\text{L}}$ to codebook embeddings $\mathbf{C}$. These embeddings condition a pretrained generative transformer through two paths: a context adapter supplies tokens to its joint-attention stream, and a time-conditioned spatial adapter injects aligned residuals into the latent input and selected transformer layers. Let $\mathbf{H}_{\text{ref}}^{(j)}$ denote the token-feature matrix at layer j of the Refiner and τ denote flow time. The layer update is

$$\mathbf{H}_{\text{ref}}^{(j+1)} = \mathcal{B}^{(j)}\left(\mathbf{H}_{\text{ref}}^{(j)}, \mathcal{C}_\phi(\mathbf{C})\right) + \mathcal{J}_\phi^{(j)}(\mathcal{A}_\phi(\mathbf{C}, \tau)), \tag{10}$$

where $\mathcal{C}_\phi$ forms global context tokens and $\mathcal{J}_\phi^{(j)} \circ \mathcal{A}_\phi$ provides spatially aligned residual features. The injection projections are zero-initialized, so the added spatial path initially preserves the backbone computation. The

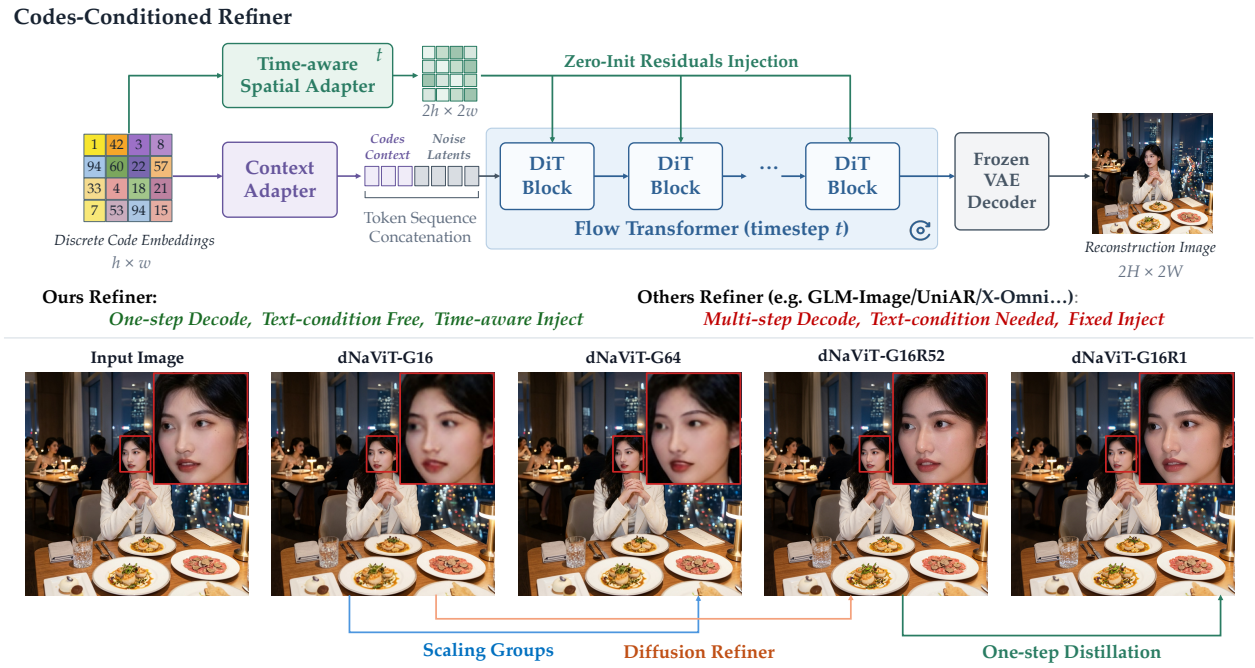


Figure 6 Overview of the code-conditioned Refiner. Code embeddings condition a flow decoder through global context tokens and time-conditioned spatial residuals. A multi-step teacher is distilled into an efficient one-step student with adversarial supervision. The timestep t in the diagram corresponds to τ in the equations. Increasing the quantization group count (G64) improves reconstruction fidelity, whereas combining fewer groups (G16) with the one-step Refiner preserves fidelity while enhancing perceptual quality.

global path carries scene-level context, while the spatial path aligns generated features with corresponding code locations. Because both adapters operate on code token grids, the Refiner supports variable resolutions and aspect ratios. The teacher is trained by flow matching in the latent space of a frozen image autoencoder. Let $\mathbf{y}_0 = \mathcal{V}(\mathbf{x}_{\mathrm{H}})$, and use the noise level $\tau \in [0, 1]$ as the flow coordinate, with $\mathbf{y}_\tau = (1-\tau)\mathbf{y}_0 + \tau\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. We optimize

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}\Big[w(\tau)\,\|\mathbf{v}_\theta(\mathbf{y}_\tau, \tau, \mathbf{C}) - (\boldsymbol{\epsilon} - \mathbf{y}_0)\|_2^2\Big]. \tag{11}$$

Here $\mathbf{v}_\theta$ is the conditional velocity field and $w(\tau)$ is the timestep-dependent loss weight.

One-step adversarial distillation. The one-step student is initialized from the frozen teacher and retains the same code condition. Training combines latent-space regression to the teacher's multi-step output with a stop-gradient correction derived from the teacher's noise predictions on identically perturbed student and teacher outputs. Adversarial supervision comes from a time-aware discriminator on features extracted by the frozen teacher, using ground-truth image latents as real samples and a hinge discriminator loss.

3 Implementation

3.1 Training Stage

Our optimization strategy separates learning complementary visual detail from adapting the enriched representation to a discrete bottleneck. As illustrated in Fig. 7, training proceeds through continuous pixel pretraining, discrete semantic reconstruction, and joint reconstruction. The tokenizer combines a pretrained Qwen3-VL ViT (∼400M), frozen throughout training to preserve its semantic representations, with a residual modulator (∼100M), totaling ∼500M parameters. We denote by $\mathcal{E}(\mathbf{x})$ the collection of intermediate and final ViT features supplied to ResMod $\mathcal{R}$, and by $\mathcal{Q}$ the grouped quantizer.

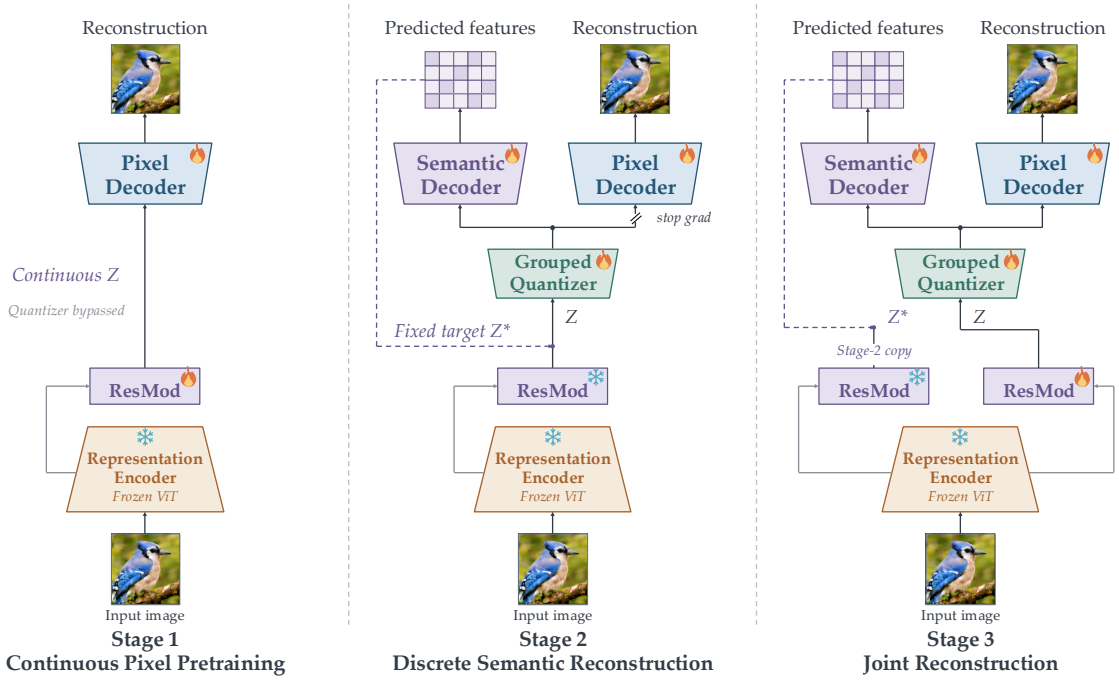


Figure 7 Three-stage training schedule of dNaViT. Continuous pixel pretraining is followed by discrete semantic reconstruction and joint reconstruction. Stage 2 stops pixel gradients at the quantized embeddings. Stage 3 opens this gradient path while a frozen copy of the Stage-2 ResMod supplies the feature target. Flame and snowflake icons mark trainable and frozen modules; dashed purple paths denote fixed targets.

Stage 1: Continuous Pixel Pretraining. We first train ResMod to recover the original visual signal from the representation encoder. ResMod produces $\mathbf{Z} = \mathcal{R}(\mathcal{E}(\mathbf{x}))$, which is passed to the pixel reconstruction branch. Minimizing $\mathcal{L}_{\text{pix}}$ jointly trains ResMod and the pixel decoder while the ViT remains frozen. This stage initializes the residual pathway to expose reconstruction-relevant information before a discrete vocabulary is introduced.

Stage 2: Discrete Semantic Reconstruction. This stage initializes discrete representations that preserve semantic structure while retaining the pixel detail introduced by ResMod in Stage 1. We freeze both the ViT and ResMod, using the ResMod output $\mathbf{Z}$ as the quantizer input and the fixed semantic reconstruction target $\mathbf{Z}^\star$. The quantizer and semantic decoder are jointly trained with $\mathcal{L}_{\text{sem}}$ to reconstruct this detail-enriched representation from $\mathbf{C} = \mathcal{Q}(\mathbf{Z})$. In parallel, the pixel decoder is trained with $\mathcal{L}_{\text{pix}}$ on detached codes $\text{sg}(\mathbf{C})$, establishing an initial mapping from discrete codes to pixels. Stopping gradients at this interface prevents the pixel objective from disrupting semantic learning while the discrete representations are being initialized.

Stage 3: Joint Reconstruction. This stage jointly refines the discrete representations for semantic and pixel reconstruction. We unfreeze the encoding ResMod while retaining a frozen teacher copy to prevent target drift during joint optimization. With student $\mathcal{R}^{\text{S}}$ and teacher $\mathcal{R}^{\text{T}}$, the semantic target $\mathbf{Z}^\star$ now comes from the frozen teacher, while the quantized features $\mathbf{C}$ come from the trainable student:

$$\begin{aligned} \mathbf{Z}^\star &= \mathcal{R}^{\text{T}}\big(\mathcal{E}(\mathbf{x})\big), \\ \mathbf{C} &= \mathcal{Q}\big(\mathcal{R}^{\text{S}}\big(\mathcal{E}(\mathbf{x})\big)\big)\,. \end{aligned} \tag{12}$$

The quantizer and both decoders continue training from Stage 2 under $\mathcal{L}_{\text{dual}}$. The semantic branch reconstructs the fixed target $\mathbf{Z}^\star$, while the pixel branch is no longer detached, allowing pixel gradients to update the quantizer and student ResMod. The ViT and teacher ResMod remain frozen throughout.

3.2 Training Data

The tokenization and de-tokenization training corpus combines diverse image sources to support semantic representation learning and maintain fine-grained pixel information. These sources are grouped into three complementary categories according to visual content. Tokenizer training uses images exclusively for dual-branch supervision, with a carefully controlled resolution distribution to support arbitrary-resolution training.

General-Domain Images. LAION (Schuhmann et al., 2022), COYO (Byeon et al., 2022), and DataComp (Gadre et al., 2023), together with images from Recap-DataComp-1B (Li et al., 2024) and CapsFusion (Yu et al., 2024b), provide broad coverage of objects, scenes, and visual styles. An in-house image corpus further expands the diversity of the training data.

Text-Rich Images and Documents. Selected subsets of TextAtlas5M (Wang et al., 2025a) include book covers, images containing long words, academic papers, and presentation slides. Chinese- and English-language book and news pages, along with additional OCR-oriented collections, extend coverage to dense text, small characters, and complex document layouts. Infographic collections and ChartGalaxy (Li et al., 2025) further contribute information-dense content, including charts and tables, where spatial relationships among visual elements are critical to interpretation.

Fine-Grained and High-aesthetic Images. Face-focused LAION subsets, images derived from SA-1B (Kirillov et al., 2023), and an in-house collection of high-aesthetic-quality images enrich the corpus with detailed facial structures, local textures, and varied image compositions.

Training begins at a fixed resolution of 256×256 using a limited set of image sources to facilitate rapid convergence, then progressively transitions to native-resolution training at resolutions up to 2K on a broader data mixture. During this transition, sampling maintains a balanced distribution across resolution ranges. Images are resized to approximately preserve their aspect ratios, align spatial dimensions with the encoder stride, and keep sequence lengths within the token budget. Across all stages, the model is trained on several hundred billion tokens.

4 Experiment

We evaluate dNaViT across five complementary tasks to assess the generality of its shared discrete representation. Image reconstruction measures the preservation of fine-grained detail, while frozen-feature classification and multimodal understanding assess semantic discriminability, transfer robustness, and visual reasoning. Autoregressive text-to-image generation and instruction-guided editing further examine the utility of the learned codes for conditional synthesis and image manipulation.

4.1 Image Reconstruction

Evaluation Setup. We evaluate reconstruction on ImageNet-1K (Russakovsky et al., 2015) and TokBench (Wu et al., 2025c) to assess whether dNaViT's shared discrete vocabulary preserves overall image fidelity and fine-grained textual and facial detail. We denote the dNaViT variants with 16 and 64 quantization groups as dNaViT-G16 and dNaViT-G64, respectively; both use a vision transformer (ViT) pixel decoder without refinement and a spatial downsampling factor of $f = 32$. dNaViT-G16-R denotes the 16-group variant with the ViT pixel decoder replaced by the code-conditioned flow decoder (Refiner; Sec. 2.6). On ImageNet-1K, we report PSNR and SSIM at 256×256, 512×512, and 1024×1024, using the full validation set and standard preprocessing. On TokBench, we report text recognition accuracy (T-ACC), normalized edit similarity (T-NED), and face similarity (F-Sim) at 512×512 and 1024×1024, with equal weighting across the official instance-scale buckets. Baseline scores are taken from published reports. Differences in evaluation protocols and training configurations preclude controlled or rate-matched comparisons.

ImageNet Reconstruction. As shown in Table 1, dNaViT-G64 achieves 25.41 dB PSNR with 0.76 SSIM at 256×256 and 28.12 dB PSNR with 0.81 SSIM at 512×512. At 1024×1024, it reaches 32.90 dB PSNR and 0.91 SSIM, compared with 28.93 dB and 0.81 for dNaViT-G16. These results indicate that dNaViT

Table 1 ImageNet reconstruction across resolutions. f denotes the spatial downsampling factor; spatial positions exclude quantization groups and do not specify bit rate. All dNaViT results use the same evaluation protocol on the full set of 50,000 ImageNet validation images. Higher is better for both metrics. Baseline results follow source-specific protocols and are not directly comparable across sources.

Method	Configuration			ImageNet	
	Resolution	f	Spatial positions	PSNR (dB) ↑	SSIM ↑
Continuous representations					
SD-VAE 1.x / 2.x (Rombach et al., 2022)	256^2	8	1024	23.54	0.68
OmniTokenizer (Wang et al., 2024a)	256^2	8	1024	26.74	0.82
SD-VAE XL (Podell et al., 2023)	256^2	8	1024	27.37	0.78
Qwen-Image (Wu et al., 2025a)	256^2	8	1024	32.18	0.90
SD-VAE 3 (Esser et al., 2024)	256^2	8	1024	31.29	0.87
Wan2.1 (Team Wan et al., 2025)	256^2	8	1024	31.34	0.89
FLUX-VAE (Black Forest Labs, 2024)	256^2	8	1024	32.74	0.92
Cosmos-CI (NVIDIA et al., 2025)	256^2	16	256	25.07	0.70
VA-VAE (Yao and Wang, 2025)	256^2	16	256	27.96	0.79
Wan2.2 (Team Wan, 2025)	256^2	16	256	31.25	0.88
PAE-DINOv2[‡] (Yue et al., 2026)	256^2	16	256	27.28	–
DC-SAE (Huang et al., 2026)	256^2	32	64	23.13	–
UniLiP (Tang et al., 2025a)	256^2	32	64	22.99	0.75
UniLiP (Tang et al., 2025a)	448^2	32	196	24.62	0.79
Discrete representations					
ARM (Wang et al., 2026)	256^2	16	256	19.60	–
Cosmos-DI (NVIDIA et al., 2025)	256^2	16	256	19.98	0.54
LlamaGen (Sun et al., 2024)	256^2	16	256	20.65	0.54
Open-MAGVIT2 (Luo et al., 2024)	256^2	16	256	22.70	0.64
BSQ-ViT (Zhao et al., 2024)	256^2	16	256	28.14	0.81
BitDance (Ai et al., 2026)	256^2	16	256	24.90	0.72
BitDance (Ai et al., 2026)	256^2	32	64	25.30	0.74
Emu3.5 (Cui et al., 2025)	512^2	16	1024	23.80	–
dNaViT-G16	256^2	32	64	22.82	0.64
dNaViT-G64	256^2	32	64	25.41	0.76
dNaViT-G16	512^2	32	256	25.47	0.71
dNaViT-G64	512^2	32	256	28.12	0.81
dNaViT-G16	1024^2	32	1024	28.93	0.81
dNaViT-G64	1024^2	32	1024	32.90	0.91

retains substantial reconstruction-relevant information within its shared discrete representation across all three resolutions.

Fine-Grained Reconstruction on TokBench. Table 2 shows that dNaViT-G64 achieves results comparable to those reported for LlamaGen-$f8$ and higher scores than DC-AE-$f32$ on all three metrics at both resolutions. At 1024×1024, it attains 63.08% T-ACC, 78.27% T-NED, and 0.67 F-Sim. The scale-stratified results in Table 3 further show consistent gains over dNaViT-G16 across all instance sizes. However, dNaViT-G16 and dNaViT-G64 differ in training configuration and duration, so this comparison does not isolate the effect of quantization group count. These findings indicate that dNaViT preserves textual and facial detail beyond global image structure. Small text nevertheless remains challenging, and a performance gap persists relative to stronger reconstruction models such as FLUX.1.

Table 2 Fine-grained reconstruction on TokBench. Results are macro-averages over three instance-scale buckets. T-ACC and T-NED are percentages; F-Sim is cosine similarity. f is the spatial downsampling factor.

Method	f	Resolution@512 × 512			Resolution@1024 × 1024		
		T-ACC↑	T-NED↑	F-Sim↑	T-ACC↑	T-NED↑	F-Sim↑
Continuous Autoencoders							
DC-AE (Chen et al., 2024a)	32	38.25	56.22	0.39	52.14	68.02	0.50
VA-VAE (Yao and Wang, 2025)	16	53.30	69.65	0.57	62.84	77.21	0.69
SD-XL (Podell et al., 2023)	8	56.86	72.55	0.55	67.84	81.15	0.70
SD-3.5 (Stability AI, 2024)	8	81.84	90.30	0.81	89.71	95.05	0.93
FLUX.1-dev (Black Forest Labs, 2024)	8	87.64	93.86	0.86	93.09	96.96	0.96
Discrete Tokenizers							
VQGAN (Esser et al., 2021)	16	6.12	17.32	0.19	15.00	28.80	0.25
Chameleon (Chameleon Team, 2024)	16	11.55	26.80	0.28	23.52	40.39	0.36
LlamaGen (Sun et al., 2024)	16	15.01	30.44	0.25	27.19	43.58	0.33
VAR (Tian et al., 2024)	16	29.31	50.00	0.32	38.36	59.25	0.40
TokenFlow (Qu et al., 2025)	16	17.40	35.49	0.24	29.30	48.09	0.31
O-MAGVIT2 (Luo et al., 2024)	16	21.65	39.63	0.31	34.74	52.03	0.40
O-MAGVIT2 (pretrain) (Luo et al., 2024)	16	27.33	47.28	0.31	39.40	58.95	0.40
UniTok (Ma et al., 2025)	16	50.10	71.08	0.44	49.70	71.59	0.53
OmniTokenizer (Wang et al., 2024a)	8	42.48	60.70	0.50	53.48	70.02	0.61
LlamaGen (Sun et al., 2024)	8	52.56	68.99	0.53	63.89	78.24	0.67
O-MAGVIT2 (Luo et al., 2024)	8	60.56	76.00	0.61	69.68	83.12	0.74
dNaViT-G16	32	33.58	52.43	0.34	45.05	62.75	0.41
dNaViT-G64	32	51.69	68.72	0.54	63.08	78.27	0.67

Table 3 Scale-stratified reconstruction with dNaViT on TokBench. S, M, and L denote the official small, medium, and large instance buckets. Text metrics are percentages. Buckets are resolution-specific and do not define a fixed-instance comparison across resolutions.

Method	Resolution	T-ACC↑			T-NED↑			F-Sim↑		
		S	M	L	S	M	L	S	M	L
dNaViT-G16	512^2	3.52	24.56	72.66	17.60	52.52	87.19	0.14	0.25	0.61
dNaViT-G64	512^2	12.18	54.94	87.95	34.73	76.67	94.76	0.29	0.49	0.84
dNaViT-G16	1024^2	10.89	38.81	85.44	28.93	65.58	93.74	0.22	0.35	0.64
dNaViT-G64	1024^2	26.79	68.50	93.95	51.93	85.29	97.58	0.49	0.65	0.86

4.2 Image Classification

Evaluation Setup. We evaluate frozen representations through linear probing on ImageNet-1K (Deng et al., 2009; Russakovsky et al., 2015) and direct transfer to ImageNet-A (Hendrycks et al., 2021b), ImageNet-R (Hendrycks et al., 2021a), ImageNet-V2 (Recht et al., 2019), ImageNet-Sketch (Wang et al., 2019), and ObjectNet (Barbu et al., 2019). All representations have a 16 × 16 spatial grid before FP32 mean pooling, followed by non-affine synchronized batch normalization and a linear classifier. Each encoder retains its original input resolution and normalization. Classifiers are trained exclusively on ImageNet-1K for 10 epochs using cross-entropy and SGD with momentum 0.9, Nesterov acceleration, zero weight decay, a global batch size of 1024, and a peak learning rate of 0.1. ImageNet-A and ImageNet-R use their respective 200-class subsets. ObjectNet evaluation covers 18,574 images from the official 113 overlapping categories, with many-to-one label mapping and red-border removal.

Table 4 STEM reasoning and general multimodal understanding. Comprehensive comparison of vision-language models (VLMs) using continuous or discrete visual representations.

Method	MMMU	MMMU-Pro	MathVista	MathVision	RealWorldQA	MMStar
Continuous						
LLaVA-OV-1.5-8B (An et al., 2025)	55.4	31.3	69.6	25.6	68.1	67.7
LLaVA-OV-2-8B (An et al., 2026b)	–	–	–	–	69.7	64.8
InternVL3-8B (Zhu et al., 2025)	62.7	–	71.6	29.3	70.8	68.2
InternVL3.5-8B (Wang et al., 2025b)	73.4	–	78.4	56.8	67.5	69.3
Kimi-VL-A3B (Kimi Team, 2025)	57.0	36.0	68.7	21.4	68.1	61.3
MiMo-VL-7B-2508 (Xiaomi, 2025)	63.1	–	–	–	–	–
BAGEL-7B (Deng et al., 2025)	55.3	–	73.1	–	–	–
Show-o2-7B (Xie et al., 2025b)	48.9	–	–	–	–	56.6
Molmo2-8B (Clark et al., 2026)	53.0	–	58.9	–	77.6	–
Nemotron-VL-12B (NVIDIA, 2025)	55.3	–	69.0	31.5	76.2	65.9
Eagle2.5-8B (Chen et al., 2025a)	55.8	–	67.8	–	76.7	66.2
Phi-4-multimodal (Microsoft, 2025)	55.1	38.5	62.4	–	–	–
PLM-8B (Cho et al., 2025)	46.1	–	–	–	75.0	–
Qwen2.5-VL-7B (Bai et al., 2025b)	58.6	38.3	68.2	25.1	–	63.9
Qwen3-VL-8B (Bai et al., 2025a)	69.6	55.9	77.2	53.9	71.5	70.9
Discrete						
Emu3-Chat-8B (Wang et al., 2024b)	31.6	–	47.6*	15.6*	57.4	46.5*
Tar-7B (Han et al., 2026)	39.0	–	–	–	–	–
ViQ-7B (Yu et al., 2026)	49.1	–	–	–	–	54.2
UniAR-8B (Peng et al., 2026)	44.3	22.7*	50.8*	–	68.5	56.7*
Kelix-8B (Ding et al., 2026)	54.1	–	76.5	–	72.1	–
LLaDA2.0-Uni (Bie et al., 2026)	50.1	34.0	68.1	26.7	66.7	64.1
dNaViT-8B	70.7	44.5	79.0	56.5	72.3	66.7

Table 5 Image classification on ImageNet-1K and distribution-shift benchmarks. Frozen representations are evaluated by linear probing on ImageNet-1K and direct transfer to ImageNet-A, ImageNet-R, ImageNet-V2, ImageNet-Sketch, and ObjectNet. All results report top-1 accuracy (%). † denotes continuous dNaViT features before quantization.

Method	f	Latent Shape	IN-1K	IN-A	IN-R	IN-V2	IN-Sketch	ObjectNet
Continuous								
QwenViT (Bai et al., 2025a)	32	16×16	80.45	60.13	65.00	70.18	46.57	56.48
InternViT (Wang et al., 2025b)	28	16×16	76.19	30.23	59.98	65.68	35.77	46.36
MoonViT (Kimi Team, 2025)	28	16×16	75.40	23.96	49.24	63.34	34.88	45.46
dNaViT-G16†	32	16×16	80.40	60.25	65.19	70.18	46.56	56.48
Discrete								
UniAR (Peng et al., 2026)	32	16×16	66.54	14.52	36.13	54.11	15.40	29.16
UniTok (Ma et al., 2025)	16	16×16	78.56	47.81	71.10	68.18	51.03	60.53
TokenFlow (Qu et al., 2025)	16	16×16	48.32	6.88	24.81	37.93	15.35	15.89
Emu3.5 (Cui et al., 2025)	16	16×16	8.39	1.16	3.08	5.87	0.77	2.71
dNaViT-G16	32	16×16	79.47	56.35	68.09	69.81	45.02	59.79

Results. As shown in Table 5, continuous dNaViT closely matches QwenViT on ImageNet-1K and all five distribution-shift benchmarks. Discrete dNaViT achieves 79.47% on ImageNet-1K, 0.93 percentage points below its continuous counterpart. Among the evaluated discrete representations, it achieves the highest

Table 6 OCR, document, and chart understanding. Comprehensive comparison of vision-language models (VLMs) using continuous or discrete visual representations.

Method	OCRBench	OCRBench V2 (en/zh)	CharXiv (DQ/RQ)	DocVQA	InfoVQA	OmniDocBench (en/zh)↓
Continuous						
LLaVA-OV-1.5-8B (An et al., 2025)	829	–	70.9/–	95.0	78.4	–
LLaVA-OV-2-8B (An et al., 2026b)	782	–	–	95.2	74.4	–
InternVL3-8B (Zhu et al., 2025)	880	–	73.6/37.6	92.7	76.8	–
InternVL3.5-8B (Wang et al., 2025b)	832	–	72.2/44.4	92.3	76.2	–
Kimi-VL-A3B (Kimi Team, 2025)	867	–	–	–	83.2	–
MiMo-VL-7B-2508 (Xiaomi, 2025)	881	–	–	–	–	–
Molmo2-8B (Clark et al., 2026)	–	–	–	93.2	80.1	–
Nemotron-VL-12B (NVIDIA, 2025)	856	62.0/44.2	76.5/41.7	94.7	79.4	–
Eagle2.5-8B (Chen et al., 2025a)	869	–	–	94.1	80.4	–
Phi-4-multimodal (Microsoft, 2025)	844	–	–	93.2	72.7	–
PLM-8B (Cho et al., 2025)	870	–	–	94.6	80.9	–
Qwen2.5-VL-7B (Bai et al., 2025b)	864	56.3/57.2	73.9/42.5	95.7	82.6	0.31/0.40
Qwen3-VL-8B (Bai et al., 2025a)	896	65.4/61.2	83.0/46.4	96.1	83.1	0.17/0.26
Discrete						
Emu3-Chat-8B (Wang et al., 2024b)	687	–	36.8*/14.1*	76.3	42.7*	–
X-Omni-7B (Geng et al., 2025)	704	–	–	88.6	–	–
ViQ-7B (Yu et al., 2026)	711	–	–	88.9	55.3	–
UniAR-8B (Peng et al., 2026)	833	43.0*/48.6*	63.3*/26.8*	91.4	68.9*	0.48*/0.43*
Kelix-8B (Ding et al., 2026)	867	–	–	–	–	–
LLaDA2.0-Uni (Bie et al., 2026)	757	–	68.4/–	89.5	70.1	–
dNaViT-8B	885	57.6/60.1	86.5/51.0	96.2	79.2	0.17/0.22

accuracy on ImageNet-1K, ImageNet-A, and ImageNet-V2, reaching 56.35% and 69.81% on the latter two. Its transfer performance is not uniformly lower than that of continuous dNaViT: it attains 68.09% on ImageNet-R and 59.79% on ObjectNet, compared with 65.19% and 56.48%, respectively. UniTok remains stronger on ImageNet-R, ImageNet-Sketch, and ObjectNet. These results suggest that dNaViT's discrete representation preserves substantial linearly accessible semantic information, although its relative transfer performance depends on the target distribution.

4.3 Visual Understanding

We evaluate whether the discrete codes learned with both semantic and pixel reconstruction retain the information needed for multimodal understanding. We report general and mathematical reasoning separately from text, document, and chart understanding.

Training Setup. We pair the frozen visual tokenizer dNaViT-G16 with Qwen3-8B and train through alignment, pre-training, mid-training, and supervised fine-tuning using approximately 70B, 150B, 60B, and 30B tokens, respectively. Alignment uses image–text pairs, while pre-training covers image captioning, interleaved image–text sequences, OCR, grounding, multi-image data, and text-only data. Mid-training focuses on STEM and OCR, supplemented with general image–text and text-only data. Supervised fine-tuning uses visual instruction-following data covering general understanding, STEM, OCR, and document tasks.

Evaluation Setup. For general understanding, we use MMMU (Yue et al., 2024), MMMU-Pro (Yue et al., 2025a), RealWorldQA (xAI, 2024), and MMStar (Chen et al., 2024b); for mathematical reasoning, we use MathVista (Lu et al., 2024) and MathVision. For fine-grained visual understanding, we evaluate OCRBench (Liu et al., 2024), OCRBench V2 (Fu et al., 2026), DocVQA (Mathew et al., 2021), InfoVQA (Mathew et al.,

Table 7 Text-to-image generation. GenEval2 GM is the GenEval2 overall score; QwenImageBench is the overall score on Qwen-Image-Bench (0–100); SpatialGenEval measures spatial-relation generation; LongText reports the English and Chinese subsets of LongText-Bench. Comp. is per-axis spatial compression, including latent packing.

Model	Comp.	Params.	GenEval2 GM	QwenImage Bench	Spatial GenEval	LongText EN	LongText CN
Proprietary Models							
GPT Image 1 (OpenAI, 2025)	–	–	–	54.07	0.61	0.96	0.62
Seedream 4.0 (ByteDance Seed, 2025)	–	–	–	56.21	0.63	0.92	0.93
Seedream 4.5 (ByteDance Seed, 2026)	–	–	–	56.78	–	0.99	0.99
GPT Image 2 (OpenAI, 2026)	–	–	–	64.69	–	0.96	0.98
Nano Banana (Google, 2025a)	–	–	44.6	–	0.62	0.87	0.33
Nano Banana 2.0 (Google Cloud, 2026)	–	–	–	59.82	–	0.98	0.95
Nano Banana Pro (Google, 2025b)	–	–	–	59.45	–	0.98	0.95
Diffusion Models							
HiDream-I1-Full (Cai et al., 2025)	16×	17B	–	43.08	–	0.54	0.02
FLUX.1 [dev] (Black Forest Labs, 2024)	16×	12B	18.3	21.58	0.57	0.61	0.01
BAGEL (Deng et al., 2025)	16×	14B	23.1	42.04	0.57	0.37	0.31
Z-Image (Z-Image Team, 2025)	16×	6B	31.8	49.10	–	0.94	0.94
Z-Image-Turbo (Z-Image Team, 2025)	–	6B	31.0	–	0.59	0.92	0.93
Qwen-Image (Wu et al., 2025a)	16×	20B	33.8	49.23	0.61	0.94	0.95
Qwen-Image-2512 (Qwen Team, 2025a)	–	20B	29.0	52.06	0.60	0.96	0.97
OmniGen2 (Wu et al., 2025b)	16×	4B	18.7	40.70	0.56	0.56	0.06
SANA-1.5 (Xie et al., 2025a)	32×	4.8B	17.5	35.14	0.54	0.05	–
SenseNova-U1-8B (Diao et al., 2026)	–	8B	40.5	49.02	0.72	0.95	0.82
Autoregressive Models							
Janus-Pro-7B (Chen et al., 2025b)	16×	7B	4.6	23.53	0.51	0.02	0.01
GLM-Image (Z.ai, 2026)	16×	16B	22.2	48.19	0.58	0.95	0.98
Emu3.5 (Cui et al., 2025)	16×	34.1B	25.4	49.89	0.57	0.98	0.93
dNaViT-8B	32×	8B	49.5	52.17	0.68	0.94	0.94

2022), ChartQA (Masry et al., 2022), CharXiv (Wang et al., 2024c), and OmniDocBench (Ouyang et al., 2025). The comparison includes models that use continuous visual representations and models that pass visual information through a discrete bottleneck before language-model processing. These are model-level comparisons with potentially different training data, image resolutions, prompting, and scoring protocols.

STEM and General Understanding. Table 4 demonstrates strong STEM reasoning and general understanding with discrete visual tokens. dNaViT leads the discrete models in this comparison and remains competitive with strong continuous-representation models, including comparable performance to Qwen3-VL-8B on MathVista. These results demonstrate that discrete visual representations can support strong multimodal reasoning and general understanding.

OCR, Document, and Chart Understanding. Discrete representations are often assumed to discard the fine-grained visual details required for OCR and document understanding (Yu et al., 2026). Table 6 shows that dNaViT outperforms the compared discrete models and substantially narrows the gap to strong continuous models on text-centric tasks. It performs comparably to Qwen3-VL-8B on DocVQA and OmniDocBench and leads the compared models on both CharXiv tasks. These results demonstrate that discrete visual representations can support fine-grained OCR, document understanding, and chart interpretation, challenging the view that they are inherently unsuitable for fine-grained visual tasks.

Table 8 Image editing on GEdit-Bench (English full set). SC: semantic consistency; PQ: perceptual quality. Higher is better for all three metrics. Comp. is per-axis spatial compression, including latent packing.

Model	Comp.	Params.	SC ↑	PQ ↑	Overall ↑
Proprietary Models					
Gemini 2.0 (Kampf and Brichtova, 2025)	–	–	6.73	6.61	6.32
Gemini 2.5 Flash Image (Fortin et al., 2025)	–	–	7.41	7.96	7.10
GPT-Image-1 [High] (OpenAI, 2025)	–	–	7.85	7.62	7.53
Open Models					
AnyEdit (Yu et al., 2024a)	8×	0.9B	3.18	5.82	3.21
Instruct-Pix2Pix (Brooks et al., 2023)	8×	0.9B	3.58	5.49	3.68
MagicBrush (Zhang et al., 2023)	8×	0.9B	4.68	5.66	4.52
UniWorld-v1 (Lin et al., 2025)	16×	12B	4.93	7.43	4.85
OmniGen (Xiao et al., 2024)	16×	3.8B	5.96	5.89	5.06
FLUX.1 Kontext [Dev] (Black Forest Labs et al., 2025)	16×	12B	6.52	7.38	6.00
OmniGen2 (Wu et al., 2025b)	16×	4B	7.16	6.77	6.41
BAGEL (Deng et al., 2025)	16×	14B	7.36	6.83	6.52
Step1X-Edit (Liu et al., 2025)	16×	12B	7.66	7.35	6.97
Qwen-Image-Edit (Qwen Team, 2025b)	16×	20B	8.00	7.86	7.56
Emu3.5 (Cui et al., 2025)	16×	34.1B	8.11	7.70	7.59
dNaViT-8B	32×	8B	7.41	7.20	7.30

4.4 Visual Generation and Editing

Evaluation Setup. We evaluate text-to-image generation of dNaViT-G16-R on GenEval2, Qwen-Image-Bench (Li et al., 2026), SpatialGenEval, and LongText-Bench (Geng et al., 2025). These benchmarks assess compositional alignment, prompt following, spatial reasoning, and text rendering in English and Chinese. For instruction-guided image editing, we use the English full set of GEdit-Bench (Liu et al., 2025) and report semantic consistency (SC), perceptual quality (PQ), and the benchmark's overall score.

Text-to-Image Generation. Table 7 compares proprietary, diffusion, and autoregressive models on compositional generation, spatial reasoning, and text rendering, using the benchmark settings summarized in its caption. dNaViT-8B attains the highest GenEval2 GM (49.5) among the models that report this metric and the second-highest SpatialGenEval score (0.68), behind SenseNova-U1-8B (0.72). Its QwenImageBench score of 52.17 exceeds every open-weight baseline while remaining below the proprietary systems (54.07–64.69), and it reaches 0.94 and 0.94 on the English and Chinese subsets of LongText-Bench, within the range of the open-weight models and below the strongest proprietary systems on Chinese text. Baseline values are reported results and may use different training data, sampling configurations, and evaluation protocols.

Instruction-Guided Image Editing. Table 8 compares image editing with proprietary and open models. SC measures consistency with the editing instruction and PQ measures perceptual quality. We retain the benchmark-reported overall scores rather than recomputing them as the arithmetic mean of SC and PQ.

5 Analysis

5.1 Effectiveness of ResMod

We compare ResMod (*a zero-initialized residual modulation module with the pretrained encoder frozen*) with SelfDistill (*a shallow fusion baseline that trains the encoder while matching its final-layer features to a frozen teacher*) Tang et al. (2025b); Yue et al. (2025b). As shown in Fig. 8, ResMod maintains lower pixel MSE and representation cosine losses during the late training stage. The zoomed 10k–11k interval shows that both

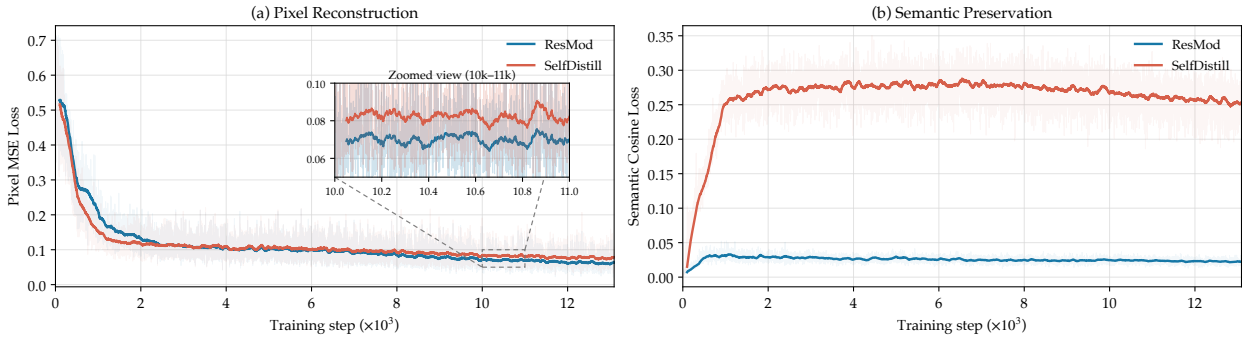


Figure 8 Pixel and representation losses for residual feature fusion. Compared with SelfDistill, ResMod maintains lower pixel MSE and representation cosine losses during late-stage training, indicating better preservation of both visual details and semantic representations.

Table 9 Visual understanding with continuous and discrete representations. Continuous features are evaluated with and without ResMod. OCRBench is divided by 10 when computing the eight-benchmark average.

Exp ID	Type	ResMod	MMMU	RealWorldQA	MMStar	OCRBench	TextVQA	AI2D	ChartQA	DocVQA	Avg. ↑
1		✗	39.44	60.26	44.60	695	74.84	**81.87**	**73.64**	**86.29**	66.31
2	Continuous	✓	**40.56**	60.92	45.33	695	74.89	81.51	73.16	86.01	66.48
3		✓	40.00	**62.61**	**46.60**	**713**	**75.05**	81.31	72.80	86.01	**66.96**
4	Discrete	✓	38.33	59.08	45.73	682	73.49	80.60	71.00	83.98	65.05

advantages remain stable rather than arising from isolated fluctuations. These results indicate that ResMod preserves high-level semantic representations while retaining more fine-grained visual details, resulting in more effective fusion of pixel and semantic information.

We further assess the effectiveness of ResMod on visual understanding tasks. These experiments use Qwen3-1.7B as the language backbone, with the visual tokenizer kept frozen. In Table 9, settings 1 and 3 train understanding models on frozen continuous features without and with pretrained ResMod, respectively. For the training-free setting 2, we directly insert pretrained ResMod into the trained model from 1 at evaluation time, without any further parameter updates. Setting 2 maintains comparable average performance to 1, even without retraining the LLM. Together with the reconstruction results above, this provides further evidence that ResMod can enrich pixel-level information while preserving pretrained semantic information.

5.2 Comparison between Discrete and Continuous Approaches

Using the same understanding training and evaluation setup as in Sec. 5.1, we compare discrete and continuous visual representations. Table 9 shows comparable performance across general understanding, OCR, and document benchmarks. This observation is consistent with the scaling analysis in LongCat-Next (Meituan LongCat Team et al., 2026), which finds that discrete visual embeddings require more training data because they are learned from scratch, and that increasing the data scale progressively closes the gap with continuous models. Together, these findings suggest that discrete visual representations can match continuous features in multimodal understanding.

5.3 Comparison of Pixel Reconstruction

Fig. 9 isolates the effect of the decoding strategy while keeping the visual representation and input resolution fixed. The ViT pixel decoder maps the discrete code embeddings directly to pixel values. This path reliably preserves the global layout, color distribution, and semantic content, but tends to over-smooth high-frequency information. Consequently, small text, fine boundaries, and repetitive textures often appear blurred or partially merged.

The flow-based Refiner, used in dNaViT-G16-R, conditions the decoder on the same discrete representations while leveraging a pretrained generative prior. Rather than relying solely on direct pixel regression, it recovers sharper letter contours, finer textures, and more convincing high-frequency details. These improvements are obtained without changing the underlying codes, and the generated results remain consistent with the structure and semantic content specified by the tokenizer.

The advantage becomes particularly evident under high spatial compression. Even with a compact token representation, the Refiner continues to recover legible text and boundary details that the pixel decoder leaves ambiguous. This indicates that the discrete representation retains substantial information beyond what is recovered by direct pixel decoding, and that a generative decoder can realize this information more effectively. The comparison therefore separates the capacity of the learned tokenizer from the limitations of the decoding head, highlighting the benefit of our code-conditioned refinement under highly compressed representations.

5.4 Qualitative Visual Understanding Examples

We present qualitative examples of our 8B understanding model on challenging visual inputs: dense full-page documents in English (Fig. 10) and Chinese (Fig. 11), cursive handwriting (Fig. 12), an engineering drawing (Fig. 13), and diagram-grounded reasoning (Figs. 14 and 15). Each example places the source image inside the model response, with the instruction and the response set. Responses are reproduced verbatim, including the model's own errors. Long transcriptions are truncated at a paragraph boundary and marked as continuing, and for examples in which the model produces an explicit reasoning trace, only the final response is shown. These examples illustrate fine-grained recognition and reasoning rather than replace quantitative evaluation.

6 Conclusion

This work presents dNaViT, a foundational visual component acting as a native-resolution tokenizer that turns discrete codes into universal visual primitives. Through lightweight residual modulation, grouped product quantization, and progressive dual-branch optimization, dNaViT successfully reconciles high-level semantic abstraction with fine-grained visual reconstruction. Further code-conditioned refinement enhances perceptual fidelity without altering the unified representation. Across understanding, generation, and editing tasks, dNaViT matches or exceeds task-specific baselines under a significantly smaller data budget. These findings demonstrate that a shared discrete token space can serve as a universal visual currency, paving the way for fully unified, native multimodal intelligence.

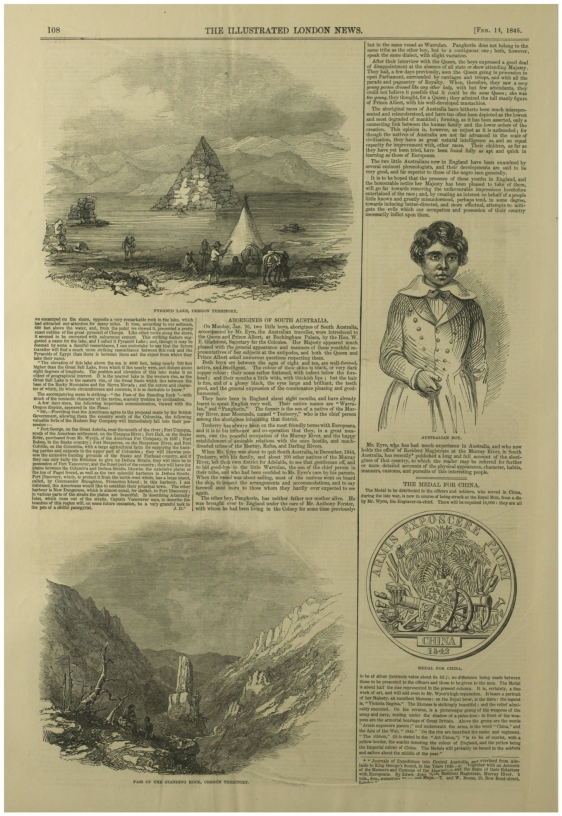
7 Contributors

Xiaotong Li[*], Zhengrong Yue[*], Xu Huang[*], Jialv Zou[*], Minhao Jing, Wenhao Yan, Haodong Li, Hai Huang, Bo Zhao, Yan Feng, Peng Pei, Xunliang Cai

[*]Equal contribution.

Figure 9 Qualitative reconstruction comparison at the same input resolution. f denotes the spatial downsampling factor. Continuous and Discrete denote unquantized latent representations and quantized codes, respectively. G and R specify the number of quantization groups and refinement steps.

Full-Page OCR of an 1846 Illustrated Newspaper



Q: Transcribe all text on this page verbatim in reading order, including headers, image captions, footnotes and the smallest print. Preserve the original language, paragraphs and punctuation. Do not summarize or translate.

A:

108
THE ILLUSTRATED LONDON NEWS.
[FEB. 14, 1846.

PYRAMID LAKE, OREGON TERRITORY.

we encamped on the shore, opposite a very remarkable rock in the lake, which had attracted our attention for many miles. It rose, according to our estimate, 600 feet above the water, and, from the point we viewed it, presented a pretty exact outline of the great pyramid of Cheops. Like other rocks along the shore, it seemed to be encrusted with calcareous cement. This striking feature suggested a name for the lake, and I called it Pyramid Lake; and, though it may be deemed by some a fanciful resemblance, I can undertake to say that the future traveller will find a much more striking resemblance between this rock and the Pyramids of Egypt than there is between them and the object from which they take their name.

"The elevation of this lake above the sea is 4890 feet, being nearly 700 feet higher than the Great Salt Lake, from which it lies nearly west, and distant about eight degrees of longitude. The position and elevation of this lake make it an object of geographical interest. It is the nearest lake to the western rim, as the Great Salt Lake is to the eastern rim, of the great Basin which lies between the base of the Rocky Mountains and the Sierra Nevada; and the extent and character of which, its whole circumference and contents, it is so desirable to know."

The accompanying scene is striking—"the Pass of the Standing Rock"—with much of the romantic character of the ravine, scarcely trodden by civilization.

A few days since, the following important communication, viewed on the Oregon dispute, appeared in the Times:

"Sir,—Providing that the Americans agree to the proposal made by the British Government, allowing them the country south of the Columbia, the following valuable forts of the Hudson Bay Company will immediately fall into their possession:—

"Fort George, on the Great Astoria, near the mouth of the river; Fort Umpqua, south of the American settlement, on the Umpqua River; Fort Hall, on the Snake River, purchased from Mr. Wyeth, of the American Fur Company, in 1837; Fort Balsee, in the Snake country; Fort Nespecees, on the Nespecees River, and Fort Colville, on the Columbia, with a large agricultural farm for supplying the hunting parties and outposts in the upper part of Columbia; they will likewise possess the extensive hunting grounds of the Snake and Flathead country, and if they can only bully the British to give up Defuca Straits, they will then be in possession of Fort Vancouver; and the finest part of the country; they will have the plains between the Columba and Defuca Straits, likewise the extensive plains at the top of Paget Sound, as well as the two splendid harbours in Defuca Straits, Port Discovery, which, to protect it from the north-west winds, has a large island, called, by Commander Broughton, Protection Island; in this harbour, I am informed, the Americans would like to establish their principal town. The other harbour is New Dungeness, which is almost equal, for shelter, to Port Discovery; in various parts of the straits the plains are beautiful. In describing Admiralty Inlet, which runs out of the straits, Captain Vancouver says, to describe the beauties of this region will, on some future occasion, be a very grateful task to the pen of a skilful panegyrist. J. D."

ABORIGINES OF SOUTH AUSTRALIA.

On Monday, Jan. 26, two little boys, aborigines of South Australia, accompanied by Mr. Eyre, the Australian traveller, were introduced to the Queen and Prince Albert, at Buckingham Palace, by the Hon. W. E. Gladstone, Secretary for the Colonies. Her Majesty appeared much pleased with the general appearance and manners of these youthful representatives of her subjects at the antipodes, and both the Queen and Prince Albert asked numerous questions respecting them.

Both boys are between the ages of eight and ten, are well-formed, active, and intelligent. The colour of their skins is black, or very dark copper colour; their noses rather flattened, with indent below the forehead; and their mouths a little wide, with thickish lips; but the hair is fine, and of a glossy black, the eyes large and brilliant, the teeth good, and the general expression of the countenance pleasing and good-humoured.

They have been in England about eight months, and have already learnt to speak English very well. Their native names are "Warrulan," and "Pangkerin." The former is the son of a native of the Murray River, near Moorunde, named "Tenberry," who is the chief person among the aborigines inhabiting that district.

Tenberry has always been on the most friendly terms with Europeans, and it is to his influence and co-operation that they, in a great measure, owe the peaceful occupation of the Murray River, and the happy establishment of amicable relations with the once hostile, and much-dreaded tribes of the Murray, Rufus, and Darling Rivers.

[response continues]

Figure 10 Full-page OCR of a densely printed page from *The Illustrated London News* (1846). The response transcribes the multi-column body text, headings and engraving captions in reading order.

Dense Chinese Academic Paper

第 22 卷第 1 期 水土保持研究 Vol. 22, No. 1
2015 年 2 月 Research of Soil and Water Conservation Feb., 2015

小兴安岭南麓典型森林类型的土壤水文功能研究

张淑兰[1]，张海军[1]，张 武[1]，王长宝[1]，邓际华[2]，宋英博[3]，张 强[1]

(1. 佳木斯大学，黑龙江 佳木斯 154007；2. 黑龙江省合江林业科学研究所，黑龙江 佳木斯 154002；3. 黑龙江省农业科学院佳木斯分院，黑龙江 佳木斯 154007)

摘 要：通过测定土壤水文物理性质，持水及渗透性能，分析了小兴安岭南麓典型森林土壤的持水、蓄水和渗透能力，并利用 Kostiakov 方程、Horton 方程和 Philip 方程模拟了土壤入渗过程。结果表明：该区森林土壤容重介于 0.45～1.52 g/cm³ 之间；不同植被类型饱和含水率为 28.73%～177.32%，毛管含水率 25.26%～156.84%，自然含水率 13.91%～119.18%，随着土层深度加厚，三者具有明显减小趋势；土壤最大持水量范围为 282.88～341.41 mm，有效持水量为 27.88～61.13 mm，涵蓄降水能力为 165.98～197.37 mm（除人工兴安落叶松林为 76.32 mm 以外）；各植被类型土壤初渗速率和稳渗速率最大值在 0—10 cm 或 10—20 cm 土层，平均值分别为 11.92 mm/min 和 2.78 mm/min，二者最小值在不同的植被类型下出现的土层不同，平均值分别为 2.19 mm/min 和 0.66 mm/min；从土壤的入渗过程模拟看，Horton 方程的模拟效果较好。

关键词：土壤水文物理；持水能力；渗透性；小兴安岭

中图分类号：S715.7 文献标识码：A 文章编号：1005-3409(2015)01-0140-06

DOI:10.13869/j.cnki.rswc.2015.01.027

Soil Hydrological Properties of Main Vegetation in Down South of Xiao Xing'an Mountains

ZHANG Shulan[1], ZHANG Haijun[1], ZHANG Wu[1], WANG Changbao[1], DENG Jihua[2], SONG Yingbo[3], ZHANG Qiang[1]

(1. *Jiamusi University, Jiamusi, Heilongjiang 154007, China*; 2. *Heijiang Forestry Institute of Heilongjiang Academy of Forestry Sciences, Jianusi, Heilongjiang 154002, China*; 3. *Jiamusi Branch of Heilongjiang Academy of Agricultural Sciences, Jianusi, Heilongjiang 154007, China*)

Abstract: Based on soil hydrology physical properties, water holding capacity and infiltration, the water storage and infiltration capacity were analyzed in typical forest stands in down south of Xiaoxing'an Mountains, and Kostiakov equation, Horton equation and Philip equation were used to simulate the soil infiltration processes. The results showed that soil bulk density was between 0.45～1.52 g/cm³, and soil saturated moisture contents of different vegetation types were between 28.73%～177.32%, capillary moisture contents were 25.26%～156.84%, the natural moisture contents were 13.91%～119.18%, and these three parameters obviously decreased with increase of soil depth; the range of soil maximum storage capacity was 282.88～341.41 mm, effective capacity was 27.88～61.13 mm, retention ability of precipitation was 165.98～197.37 mm (except for artificial *Larix gmelini* forest of 76.32 mm). Soil initial infiltration rate and steady infiltration rate of each vegetation type were the largest in 0—10 cm or 10—20 cm soil layer, the average value was 11.92 mm/min and 2.78 mm/min, respectively, and their minimum values in different vegetation types present in different soil layers, with average value of 2.19 mm/min and 0.66 mm/min. According to the soil infiltration process simulation, Horton equation was better for this simulation.

Keywords: soil hydrological properties; water storage capacity; infiltration; Xiao Xing'an Mountains

森林土壤的水文功能主要指土壤的持水能力及渗透特性，是森林植被保持水土和涵养水源作用的重要体现[1-3]，也是森林生态系统服务功能评价的一个重要指标[4-5]。其中，森林土壤的持水能力是指森林生态系统中涵蓄土壤水分、补充地下水和调节河川流量的功能，其与森林土壤的物理性质有直接的关

收稿日期：2014-04-11 修回日期：2014-04-29

资助项目：国家自然科学基金项目（41301034）；黑龙江省教育厅科学技术研究项目（12531703）；佳木斯大学青年基金项目（Sq2012-32）

第一作者：张淑兰（1980—），女，河北省滦南县人，博士，讲师，主要从事森林生态、生态水文过程与模型应用研究。E-mail:zhangshulan1980@163.com

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A:

第 22 卷第 1 期
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Vol. 22, No. 1
Feb., 2015
小兴安岭南麓典型森林类型的土壤水文功能研究
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Soil Hydrological
Properties of Main Vegetation in Down South of Xiao Xing'an Mountains
ZHANG Shulan [1] , ZHANG Haijun [1] , ZHANG Wu [1] , WANG Changbao [1] , DENG Jihua [2] , SONG Yingbo [3] , ZHANG Qiang [1]
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第一作者：张淑兰（1980—），女，河北省滦南县人，博士，讲师，主要从事森林生态、生态水文过程与模型应用研究。E-mail:zhangshulan1980@163.com
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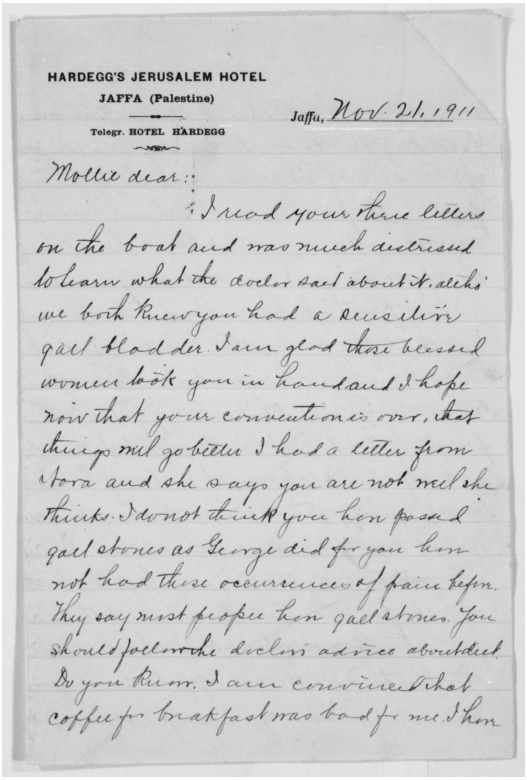
Figure 11 Full-page OCR of a dense Chinese research article, covering the bilingual header, title, author affiliations with superscripts, abstract and body text.

Handwritten Letter: Full Page


HARDEGG'S JERUSALEM HOTEL
JAFFA (Palestine)
Telegr. HOTEL HARDEGG

Jaffa, Nov. 21, 1911

Mollie dear::

I read your three letters on the boat and was much distressed to learn what the doctor said about it, also we both knew you had a sensitive gall bladder. I am glad those blessed women took you in hand and I hope now that your convalescence is over, that things will go better. I had a letter from Mora and she says you are not well she thinks. I do not think you have passed gall stones as George did for you from not had three occurrences of pain before. They say most people have gall stones. You should follow the doctor's advice about diet. Do you know. I am convinced that coffee for breakfast was bad for me. I am

Q: Transcribe all handwritten text on this page verbatim, line by line, preserving the original spelling, punctuation and line breaks. Do not correct or translate. Mark illegible words as [illegible].

A:

HARDEGG'S JERUSALEM HOTEL
JAFFA (Palestine)
Telegr. HOTEL HARDEGG
Jaffa, Nov. 21, 1911
Mollie dear::
I read your three letters
on the boat and was much distressed
to learn what the doctor said about it, also
we both knew you had a sensitive
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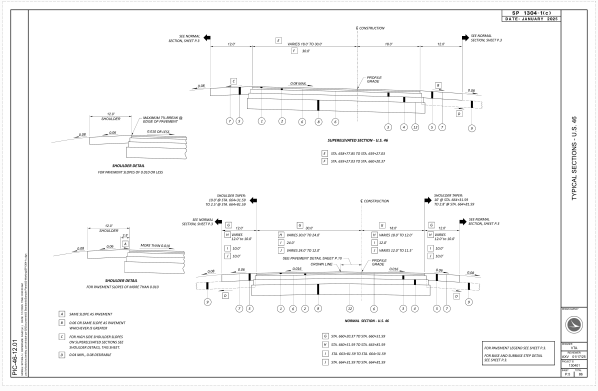
gall stones as George did for you from
not had three occurrences of pain before.
They say most people have gall stones. You
should follow the doctor's advice about diet.
Do you know. I am convinced that
coffee for breakfast was bad for me. I am

Figure 12 Full transcription of a 1911 cursive letter written on printed hotel stationery, covering both the printed letterhead and the connected handwriting.

Engineering Drawing

Q: Read the title block and every station range on this engineering drawing. Transcribe the drawing number, date, sheet title, project ID, sheet number and total, and list each STA. range under the section it belongs to. Output only the transcription.



A:

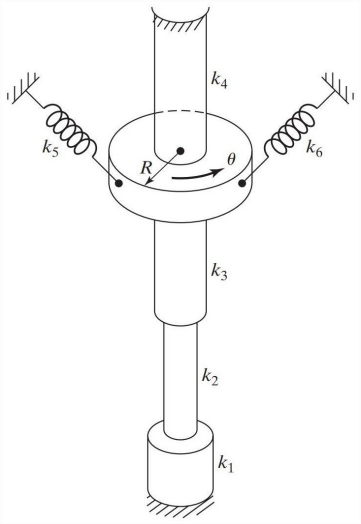
Drawing Number: SP 1304-1(c)
Date: JANUARY 2025
Sheet Title: TYPICAL SECTIONS - U.S. 46
Project ID: 130401
Sheet Number / Total: P.5 / 86

SUPERELEVATED SECTION - U.S. 46
STA. 658+77.85 TO STA. 659+27.03
STA. 659+27.03 TO STA. 660+20.37

NORMAL SECTION - U.S. 46
STA. 660+20.37 TO STA. 660+51.59
STA. 660+51.59 TO STA. 663+81.59
STA. 663+81.59 TO STA. 664+31.59
STA. 664+31.59 TO STA. 664+81.59

Figure 13 Structured field extraction from a large-format engineering drawing: the response transcribes the title block and every station range, grouped by the typical section it belongs to.

Equivalent Stiffness of a Spring System



Q: Find the equivalent torsional spring constant of the system shown in the figure. Assume that k1, k2, k3, and k4 are torsional and k5 and k6 are linear spring constants. Options: A. $k_{et} = (\frac{k_1k_2k_3}{k_2k_3+k_1k_3+k_1k_2}) + k_5 + R^2(k_4 + k_6)$; B. $k_{et} = (\frac{k_1k_2k_3}{k_2k_3+k_1k_3+k_1k_2}) + k_4 + R^2(k_5 + k_6)$; C. $k_{et} = (\frac{k_1k_2k_3}{k_2k_3+k_1k_3+k_1k_2}) + k_6 + R^2(k_4 + k_5)$.

A:

The correct choice is **B**.

The three torsional springs k_1, k_2, k_3 are in series, so their equivalent stiffness is

$$k_{123} = \frac{k_1k_2k_3}{k_1k_2 + k_1k_3 + k_2k_3}.$$

The linear springs k_5 and k_6 act at radius R, so they produce torques

$$M_5 = k_5R^2\theta, \qquad M_6 = k_6R^2\theta,$$

giving total linear-spring torque

$$M_{56} = R^2(k_5 + k_6)\theta.$$

The torsional spring k_4 directly adds

$$M_4 = k_4\theta.$$

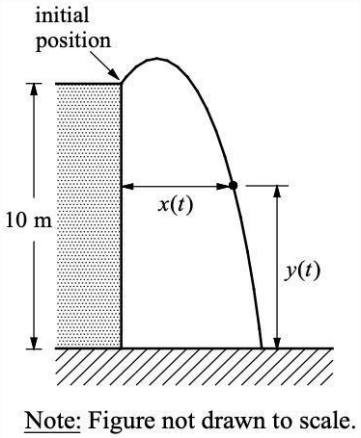
Thus

$$\boxed{k_{et} = \frac{k_1k_2k_3}{k_1k_2 + k_1k_3 + k_2k_3} + k_4 + R^2(k_5 + k_6)}$$

which is **Option B**.

Figure 14 Mechanical-vibration reasoning from a schematic: the model identifies series, parallel and lever-arm spring combinations (MMMU, hard).

Projectile Angle at Impact



Q: An obiect is thrown upward into the air 10 meters above the ground. The figure above shows the initiaposition of the object and the position at a later time. At time t seconds after the object is thrown upwardthe horizontal distance from the initial position is given by x(t) meters, and the vertical distance from the ground is given by y(t) meters, where $\frac{dx}{dt} = 1.4$ and $\frac{dy}{dt} = 4.2 - 9.8t$,for t ≥ 0 .Find the angle θ, $0 < \theta < \pi/2$, between the path of the object and the ground at the instance the objecthit the ground. Options: A. $\theta = 0.524$; B. $\theta = 1.047$; C. $\theta = 1.475$; D. $\theta = 1.570$.

A:

$$y(t) = 10 + 4.2t - 4.9t^2$$

The object hits the ground when $y(t) = 0$:

$$4.9t^2 - 4.2t - 10 = 0$$

$$t \approx 1.92$$

At that time,

$$\frac{dy}{dt} = 4.2 - 9.8t \approx -14.616$$

and

$$\frac{dx}{dt} = 1.4$$

So the slope of the path is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \approx \frac{-14.616}{1.4} = -10.44$$

The angle with the ground is

$$\theta = \tan^{-1}(10.44) \approx 1.475$$

Answer: C. $\theta = 1.475$

Figure 15 Calculus reasoning grounded in a diagram: the model integrates the velocity, solves for the landing time and converts the slope into an angle (MMMU, hard).

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